MILITARY SPECIFICATION

PEDESTAL GROUP, ANTENNA, FOR IFF ANTENNAS
AN/UPA-57

1 SCOPE

1 1 This specification covers the design and construction of a lightweight Antenna Pedestal Group AN/UPA-57 to support and position an IFF interrogator antenna, either in synchronism to a remote command input, or in synchronism to a manual command input, or independently at a selectable speed The pedestal shall operate at speeds from 2 to 30 revolutions per minute (rpm) when slaved and shall operate at speeds up to 15 rpm when free running It is intended that this IFF antenna positioning system will represent the latest advances in the field of shipboard pedestal and solidstate control design

2 APPLICABLE DOCUMENTS

2 1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein

SPECIFICATIONS

MILITARY

MIL-T-27	-	Transformer and Inductor (Audio, Power, and High Power Pulse) General Specification for
MIL-S-901	-	Shock Tests, H I (High Impact), Shipboard Machinery Equipment and Systems, Requirement for (Navy)
MIL-Q-9858	-	Quality Program Requirements
MIL-E-15090	-	Enamel, Equipment, Light-Gray (Formula No 111)
MIL-E-16400	-	Electronic Equipment, Naval Ship and Shore, General Specification (Navy)
MIL-E-17555	-	Electronic and Electrical Equipment, Accessories and Repair Parts, Packaging and Packing of
MIL-M-17556	-	Motor, Direct Current, Fractional Hp (Shipboard Use)
MIL-C-45662	-	Calibration System Requirements
MIL-P-55110	-	Printed Wiring Boards

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-108	-	Definition of and Basic Requirements for Enclosures for Electric and Electronic Equipment
MIL-STD-167	-	Mechanical Vibrations of Shipboard Equipment
MIL-STD-275	-	Printed Wiring for Electronic Equipment
MIL-STD-454	-	Standard General Requirements for Electronic Equipment
MIL-STD-461	-	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-470	-	Maintainability Program Requirements (for Systems and Equipments)
MIL-STD-756	-	Reliability Prediction
MIL-STD-781	-	Reliability Tests Exponential Distribution
MIL-STD-785	-	Reliability Program for Systems and Equipment Development and Production

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MIL-STD-1472 - Human Engineering Design Criteria for Military Systems, Equipment and Facilities

PUBLICATIONS

MILITARY

MIL-HDBK-216 - RF Transmission Lines and Fittings
MIL-HDBK-217 - Reliability Stress and Failure Rate Data for Electronic Equipment.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2 2 Other publications The following document forms a part of this specification to the extent specified herein Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposals shall apply

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, U S Government Printing Office, Washington, D C 20402)

3 REQUIREMENTS

3 1 General requirements The equipment shall be in accordance with MIL-E-16400, except as hereinafter specified Human engineering aspects of equipment and facility design shall conform to MIL-STD-1472

3 2 First article sample Prior to beginning production, a sample shall be tested as specified in 4 3 (see 6 3)

3 3 Material Screws and screw thread devices shall be in accordance with the requirements of Handbook H28

3 3.1 Aluminum or any other lightweight material acceptable to the command or agency concerned may be used where the weight of units can be significantly reduced without penalizing strength, machining accuracy, or resistance to wear and corrosion Aluminum may be used in motor construction Use of magnesium will not be permitted

3 3.2 All external hardware shall be made of austenitic corrosion-resistant steel RF transmission lines and fittings shall be in accordance with MIL-HDBK-216

3 4 Reliability

4 1 Quantitative reliability requirement The specified MTBF (as defined by MIL-STD-781) of this equipment shall be 4000 hours

3 4 2 Reliability program plan The contractor shall submit a reliability program plan in accordance with and in paragraph sequence with MIL-STD-785 (see 6 2 2)

3 4 3 Reliability prediction The contractor shall submit the following reliability predictions (see 6 2 2)

(a) Feasibility Prediction shall be made in accordance with the requirements of MIL-STD-756
(b) Design prediction based on stress factors and part populations in accordance with the requirements of MIL-HDBK-217

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3 4 4 Source of failure rates. The primary source of failure rate data shall be MIL-HDBK-217 For items not included in MIL-HDBK-217, other sources of data may be used with the approval of the command or agency concerned When such alternate sources of data are proposed, enough information shall be submitted to enable the command or agency concerned to evaluate the validity and applicability of the results derived therefrom.(see 6 2 2)

3 4.5 Reliability reports Reports in accordance with MIL-STD-785 and also providing a complete accounting of progress on each element defined by the reliability program plan, results achieved and status of actions to resolve problems and correct weak links shall be submitted to the command or agency concerned (see 6.2.2)

3 5 Design requirements

3 5 1 Composition A complete equipment shall consist of the following units

(a) Antenna pedestal assembly (see 3 7)
(b) Control power supply unit (see 3 8)
(c) Manual antenna positioning control (see 3 9)
(d) Drive motor disconnect mast switch (see 3 10)

3 5.2 Location of units Above deck units shall consist of those specified in 3 5 1, Items (a) and (d) The remaining units shall be designed for below-deck operation and installation Separation of pedestal and control power supply may be up to 1000 ft. between control power supply and manual control by up to 100 feet Normal operation shall take place under these conditions

3 5 3 Shock, vibration and inclination

3 5 3 1 Shock requirements shall be Grade A, Class 1, medium weight, type A in accordance with MIL-S-901

3 5 3.2 The vibration requirements shall be Type I of MIL-STD-167

3 5 4 Enclosure The above-deck units shall conform to the watertight enclosure requirements of MIL-STD-108 The below-deck units shall conform to the dripproof enclosure requirements of MIL-STD-108

3 5 5 Service reliability The equipment shall be designed to operate as specified herein, under conditions encountered aboard ship

3 5.6 Interconnecting cable The conductors carrying current exceeding 0 5 ampere and those conductors connected to commutated sources shall be run in cables separate from other conductors All terminal boards and cable entrances shall be designed and arranged to accommodate this division of circuits Shielded cables shall be employed to eliminate any effects of stray signal pickup on pedestal control and operation

3 5 7 Special test equipment No special test equipment or tools shall be required for maintenance or alignment of the pedestal group

3 5.8 Synchros Synchros shall be furnished by the contractor and shall be in accordance with MIL-E-16400

3 5 9 Transformers Transformers shall conform to grade 4, Class R, life expectancy S of MIL-STD-454 as applicable

3 5 10 Modular construction Potted or nonrepairable circuit boards and modules shall not be used

3 5 11 Test points All points required for trouble-shooting, aligning and maintaining the pedestal group shall be provided and shall be readily accessible

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3 5 12 Indicator lamps Indicator lamps for circuit functions such as "Power-On", shall not be of the same type as those used for fuse alarms

3 5.13 Fuses Circuit breakers shall be provided for all input power lines and all other lines on which the possibility of overload exists

3.5.13.1 Blown fuse indicators Blown fuse indicators are not applicable

3.5.14 Size and weight. The size and weight of the IFF pedestal group shall be kept to a minimum The pedestal weight shall not exceed 125 pounds The control unit weight shall not exceed 125 pounds

3.5 15 Input power. Operating power for the system shall be derived from the 220 volt (V), 3-phase, 60-hertz (Hz), and 115 V, single phase 60-Hz main bus The performance of the equipment shall meet the requirements of this specification when the input frequency varies ±10 percent from the nominal values Transient frequency operation per MIL-E-16400 is not required

3 5.16 Motors Motors shall be in accordance with MIL-M-17556 Provision shall be made for motor replacement without removal of other major parts

3 5 17 Electromagnetic interference emission, susceptibility, and equipment and systems self-compatibility The antenna shall operate within the electromagnetic interference emission, susceptibility, and equipment and systems self-compatibility limits in accordance with MIL-STD-461

3 5.18 Environmental conditions. The equipment shall be capable of operation under the following conditions

(a) Temperature Class 1 for above deck units, except the upper limit is 125°C[1]
Class 4 for below deck units, except the upper limit is 65°C[2]
(b) Humidity up to 95 percent relative humidity
(c) Wind Speed
(d) Icing
(e) Nuclear air blast (Calculations only) (3 5 pounds per square inch (psi))

3 6 Functions The functional block diagram indicating the system requirements shall be as shown on figure 1.

3 6.1 Operation The antenna pedestal shall be capable of supporting and rotating either the AS-2188 or AT-352/UPA-22 IFF antennas as follows

(a) Capable of rotating from 2 to 30 rpm when slaved to a single speed master synchro, or
(b) Capable of running freely at continuously controlled speeds of from 2 to 15 rpm, or
(c) Capable of positioning from a manual position control

3.6.1 1 Operating conditions The antenna pedestal shall be capable of operating in accordance with the above requirements in wind velocities of up to 75 knots relative, with antenna icing of up to 4 5 pounds per square foot (psf) of surface area under the environmental conditions as specified in 3 5 18

3 6 1.2 Time meters Time meters are not applicable

3 6.1 3 Rotation Rotation shall be clockwise as viewed from above, but provision shall be made so that no damage shall result if the antenna is rotated counter-clockwise, either by accident or design The rotating speed, in Rpm, shall be shown on an indicator located on the Control Power Supply Unit, together with an easily adjustable control to regulate the rotation speed in the free running condition

3 6 1 4 The Control Power Supply Unit shall be capable of positioning the pedestal as specified in 3 6 1 with a position error no greater than 1.5 degrees Position data will be supplied from a single speed synchro Provision shall be made to prevent the pedestal from running away when synchro input signal is lost

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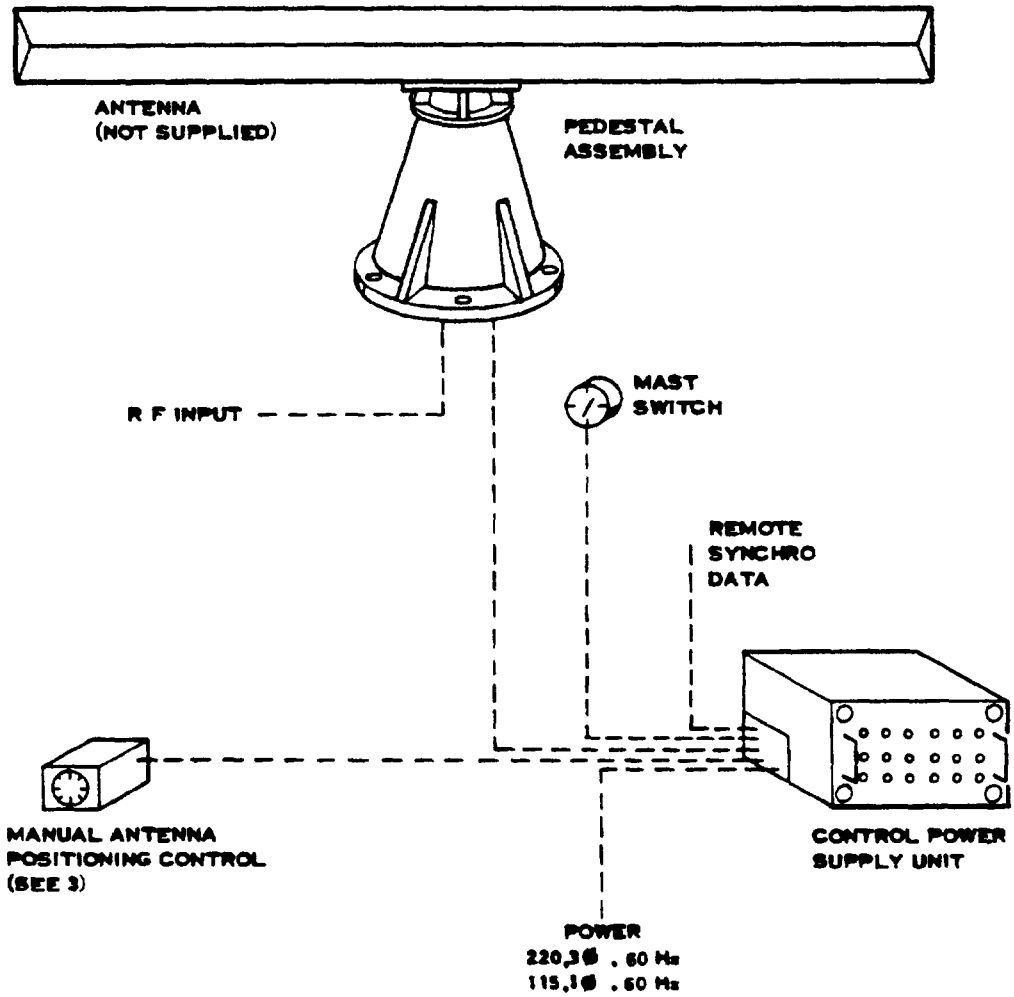


FIGURE 1 Pictorial view of antenna pedestal group

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3 6.1.5 Stowing of antenna. A stow switch shall be provided on the Control Power Supply Unit such that when this switch is closed, the antenna shall rotate to a stow position predetermined by an adjustable setting, either on the antenna control or on the pedestal, and lock in that position A manual brake release shall be provided on the pedestal Brake operation shall not be so abrupt as to damage either the pedestal or the antenna. Brake operation shall occur when primary power is lost

3 6.1 6 Stable operation The pedestal shall reach stable operation in the free-running condition within 5 minutes or less

3.6 2 Antenna Survival The Antenna Pedestal Group shall be capable of surviving 100 knot winds relative to the pedestal, with 7 psf of ice, under the environmental conditions specified in 3 5 18

3 7 Antenna pedestal assembly The antenna pedestal shall be designed to conform to the requirements of 3 6 1

3 7.1 Pedestal base The pedestal assembly shall have the necessary base support with a deck mounting flange A marker on the base shall indicate ship's foremounting The bottom of the pedestal base shall be closed to prevent the entrance of moisture or dirt

3 7.2 Pedestal gear box A modular drive gear box shall be supplied mounted on the above base containing the synchros with a backlash limit of 5 minutes of arc The drive gear train backlash shall not exceed 10 minutes of arc at the output Consideration shall be given to the use of helical gears in the design

3 7.3 Synchro transmitters A 1 1 and a 36 1 synchro transmitter shall be provided to supply azimuth information to an indicator when the pedestal is in the free running condition

3 7.4 IFF rotary joint A capacity-type IFF rotary joint, together with a slip ring capable of transmitting d c voltage and current shall be provided The joint shall be mounted on the pedestal assembly and shall be removable without disturbing the antenna or any other major group parts The IFF rotary joint shall accommodate IFF Transmissions of 15 kilowatt (kw) peak power having a duty Hz of one percent in the frequency band of 1010 to 1110 megahertz (MHz) The lower portion of the rotary joint shall have a waterproof fitting to mate with a UG-154/U connector and the upper portion shall have a waterproof fitting to mate with a UG-59B/U connector The VSWR of the rotary joint as measured at the base of the pedestal, shall not exceed 1 5 1 over the band covered herein and over 360 degrees of pedestal rotation

3 7 5 Pedestal turntable A mounting turntable which will accommodate the antenna support bracket shall be supplied The pedestal mount assembly shall be designed so that it is possible to bolt and secure the antenna from the bottom See figure 2 for details of antenna mounting brackets

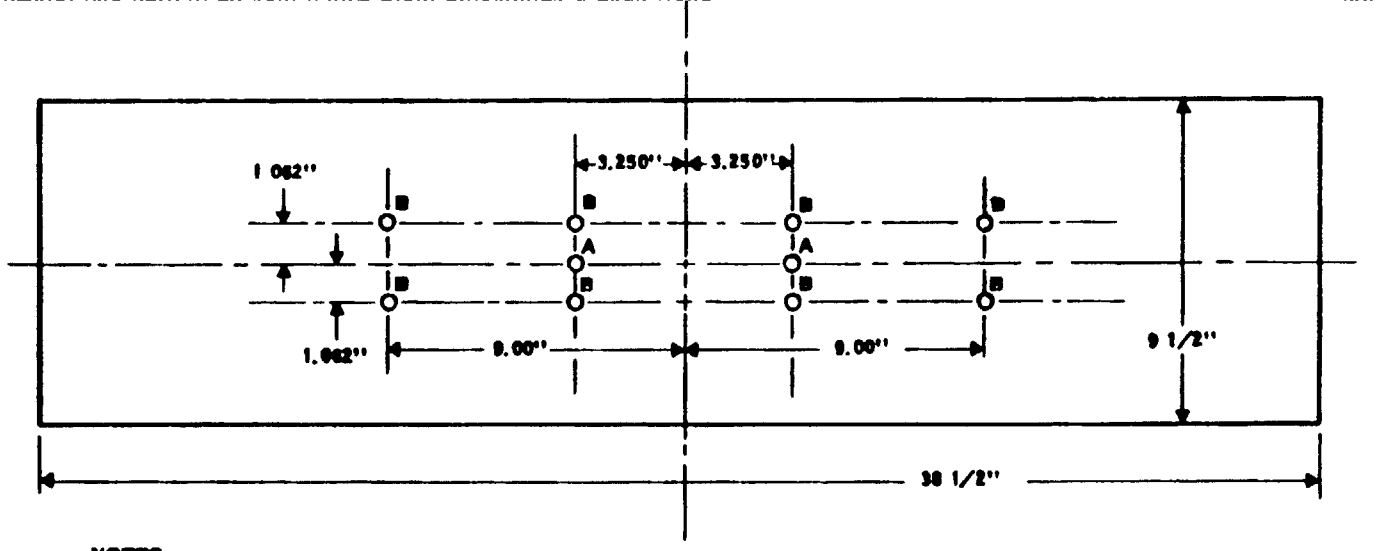
3 7 6 Ship's heading marker A ship's heading marker assembly shall be supplied consisting of a microswitch and cam assembly to provide a momentary opening or closing contact when the antenna is trained zero degree relative Provisions shall be made to adjust the ship's heading marker ±180 degrees in azimuth

3 7 7 Cable entrances Provisions shall be made for a cable entrance to the antenna pedestal assembly designed to accommodate all interconnecting cables, except radio frequency cables or lines The entrance shall have a removable plate which may be drilled and tapped at the time of installation

3 7.8 Access covers All covers which are normally removed from the antenna pedestal assembly during routine maintenance or adjustment shall be secured to the main part of the pedestal Covers which individually weigh less than 3 pounds may be secured by means of a length of chain

3 7 9 Lubrication The bearings and gears shall be lubricated by a positive method which does not depend upon the motion of the pedestal Grease lubrication is permitted Adequate access shall be supplied for lubrication maintenance If oil is used, thermostatically controlled heaters shall be supplied with remote oil level indication in the Control Power Supply Unit

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NOTES –

A. GUIDE PIN HOLES-0.3906"D (2)
B. 3/8" X 24 CAPTIVE SELF LOCKING NUTS (8)
C. HOLE TOLERANCE- ± 1/64" (0.0156")

FIGURE 2 Mounting bracket - directional IFF antenna (typical)

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3 7 10 Pedestal enclosure The pedestal shall be so designed as to reduce the intake of external air to a minimum. Provision shall be made for plugged drain holes at all low portions of the pedestal to drain atmospheric condensation

3 8 Control power supply unit The control power supply unit shall include all necessary parts for the control of the antenna system with the exception of those items which by necessity must be located elsewhere This unit shall consist of control circuitry for the automatic positioning of the antenna and all data distribution for the antenna control system and shall have "Power On" and error indicators, fuseholders, and an on-off switch readily visible and accessible A voltmeter together with a wafer switch shall be provided whereby alignment shall be readily accomplished Control locking devices are not required on this unit The unit shall be designed for hard mounting on the base only All electrical connections shall be from the rear

3 8 1 Servo control amplifier The control power supply unit shall contain the servo control amplifier Gain and anti-hunt controls shall be provided The error signal amplifier driving the pedestal drive motor power amplifier shall not contain any vacuum tubes Transistors and other semiconductor devices shall be used for the entire servo electronic circuitry

3 8.2 Power supplies All necessary power supplies for servo amplifier and motor field shall be supplied No rotating parts or vacuum tubes shall be used Semiconductor or magnetic amplifier devices shall be used where applicable

3 8.3 Primary power line voltage and frequency Primary power line voltage and frequency is not applicable

3.8.4 Accessibility Provision shall be made for turning the unit on its horizontal axis ±90°, with locking provisions at 0°, ±45° and ±90° Sufficient cable length shall be provided within the chassis to permit operation and trouble-shooting in any and all of the above positions, the cable being supported by a horizontal spring-loaded arm or similar device approved by the command or agency concerned The front panel layout, with its controls and test points, in accordance with MIL-STD-1472 shall be submitted for the approval of the command or agency concerned

3 8.5 Error indicator When the actual antenna pedestal bearing and the position signal input deviate greater than ±2 degrees, an error indicator shall be de-energized This indicator shall be of the fail safe type and shall be identified as "Error" and shall be mounted on the control power supply unit

3 9 Manual pedestal control A manual pedestal control shall be provided, as an option, to enable the operator to position the IFF antenna to any azimuth setting from a remote position It shall have a dial showing the relative antenna bearing and a control to switch from slaved operation, to free running operation, or to manual operation The performance of the system in the slaved or the free running operation shall not be impaired in any way should the manual pedestal positioning control be disconnected It shall be designed to mount on top, bottom, or either side, as desired

3 10 Pedestal disconnect mast switch A pedestal disconnect mast switch shall be provided for each pedestal installation The disconnect switch shall be designed to remove all voltage from the pedestal, with provisions for locking in the open position (Power OFF)

4 QUALITY ASSURANCE PROVISIONS

4 1 Responsibility for inspection Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements

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4 1 1 Contractor's quality assurance system The supplier shall provide and maintain an effective inspection and quality control system acceptable to the Government covering the supplies under the contract The quality control system shall be in accordance with MIL-Q-9858 A quality program plan, as it would apply to the requirements of this contract, shall be prepared prior to first article inspection (see 6 2 2) If first article inspection is not required, the quality program plan shall be prepared prior to production inspection This program plan shall include flow charts showing the complete flow of material from inspection on receipt, through all manufacturing processes, to final shipment The various processes and inspection positions shall be clearly identified on the flow-charts and cross referenced to the detailed inspection procedures and criteria of the written description of the quality control system Any changes to the quality control system or the quality program plan shall be submitted to the cognizant Government Inspector for approval prior to use.

4 1 2 Government verification All quality assurance operations performed by the contractor will be subject to Government verification at any time Verification will consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the written system description are being properly applied, (b) Government product inspection to measure quality of product to be offered for acceptance, and (c) Government product inspection of delivered products to assure compliance with all requirements of this specification Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated

4 1.3 Test procedures The supplier shall submit written test procedures including test equipment and tolerance limits to be used for first article inspection and quality conformance inspection Each test shall be identified in accordance with 4.2 as to classification (see 6 2 2)

4 1.3 1 Procedures for inspections, tests, demonstrations, and so forth, shall contain step by step information which will enable the operator to properly perform the requirements of this specification The method of examination shall be clearly defined

4 1.3.2 The procedures shall contain a list of test equipment identified by name and model Special test equipment shall also be listed and identified Test equipment shall be calibrated in accordance with MIL-C-45662

4 1 3 3 Data sheets shall be included in the procedures Conformance parameters shall be clearly indicated with tolerance limits Whenever quantitative measurements are made the result shall be shown by quantitative data

4 2 Classification of inspection The inspection requirements specified herein are classified as follows

(a) First article inspection (see 4 3)
(b) Quality conformance inspection (see 4 4)
    (1) Production inspection (see 4 4 1)
    (2) Production control inspection (see 4 4 2)
    (3) Environmental tests (see 4 4 3)
(c) Inspection of preparation for delivery (see 4 6)

4 3 First article inspection Unless otherwise specified (see 6 3), all units shall be required for first article inspection First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification First article inspection shall include the tests specified in table I

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TABLE I. First article inspection

Examination or test	Test method paragraph	Number of sample units to be inspected	Number of failures
Production inspection	4.4.1	all	0
Production control inspection	4.4.2	all	0
Environmental tests	4 4.3	all	0
Wind speed and icing tests	4.5.9	all	0
Magnetic test	MIL-E-16400	all	0

4.3 1 First article inspection report. A first article inspection report shall be prepared containing all the results of the first article inspection (see 6.2.2). The report shall include completed Form NAVEXOS 3373 as specified in MIL-S-901 for the shock test and resilient mount data.

4.4 Quality conformance inspection.

4.4 1 Production inspection Production inspection shall be made on every equipment offered for delivery. The inspection shall comprise such examination and testing as will prove the workmanship and reveal the omissions and errors of the production process such as functional and performance tests at a limited number of points in the required range, tests which detect deviations from design, tests of adjustments, and tests which detect hidden defects of material. Production inspection shall consist of the tests approved in accordance with 4 1.3 and shall include the requirements of table II.

TABLE II. Production inspection

Examination or test	Test method paragraph
Visual and mechanical examination	4.5.1
Operating test	4 5.2

4 4.2 Production control inspection. Production control inspection, including sampling, shall conform to table III below and to the procedures for special inspection levels of MIL-STD-105 The inspection level shall be S-3 for normal, tightened, and reduced inspection utilizing an AQL of 6.5 percent Production control inspection shall be performed on equipments that have passed production inspection Production control inspection shall be approved in accordance with 4.1.3 The equipment shall satisfactorily meet the requirements of production control inspection prior to release for shipment

TABLE III Production control inspection

Examination or test	Reference
Weights and dimensions	4 5 3
Supply line voltage and frequency	4 5 4
Heat test	MIL-E-16400
Enclosure test	4 5 5
Controls and control circuits	MIL-E-16400
Weld test	MIL-E-16400
Power	MIL-E-16400

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4 4 3 Environmental tests Environmental tests shall be as listed in table IV and shall be performed on units which have been subjected to and passed production inspection Environmental tests shall be approved in accordance with 4.1 3.

TABLE IV Environmental tests

Examination or test	Reference
Temperature	4 5 6
Humidity	4 5.7
Shock, vibration and inclination	4.5 8
Salt spray test	MIL-E-16400

4 4 3.1 Environmental test samples Samples for the environmental inspection shall be randomly selected in accordance with the following.

(a) One sample from the first 25 production units fabricated
(b) One sample from each successive 75 units produced, or fraction thereof

4 4 3.2 Nonconforming production control or environmental sample units If a sample unit fails the test specified in 4 4.2 or 4 4.3, the supplier shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken to correct units of product which were manufactured with the same conditions, materials, processes, and so forth If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer

4.4.3.3 Reinspection of conforming production control and environmental sample units. Unless otherwise specified in the contract, sample units which have been subjected to and passed production control and environmental tests may be accepted on the contract provided they are resubjected to and pass production inspection after repair of all damage

4 5 Inspection procedures

4 5 1 Visual and mechanical examination Each antenna pedestal group shall be examined in accordance with the workmanship criteria defined in MIL-STD-454 (requirement 9) to determine that the materials, parts, finish, construction, and workmanship requirements specified in section 3 of this specification have been met. Particular attention shall be given the following

(a) Assembly and fit, mechanical safety, and marking
(b) Treatment for prevention of corrosion
(c) Accessibility of components, parts, and test points for servicing and trouble shooting
(d) Check of pedestal mounting and hoisting means
(e) Check of antenna mounting means
(f) Check of lubrication and rust prevention
(g) Ease of operation of gears, adjustable and sliding parts, thumb screws, controls and switches
(h) Check of solder connections
(i) Cable runs between units including plugs and receptacles
(j) Completeness

4 5.2 Operating test The antenna pedestal group, with a government furnished AS2188 or AT-352 antenna installed, shall be aligned and subjected to an operating test in accordance with MIL-E-16400 except

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that the equipment shall be operated continuously for a period of 24 hours. Operating power during this test shall be in accordance with 3.5.15. The following requirements shall be verified during the operating test:

(a) Synchro backlash and drive gear train backlash check (see 3 7.2)
(b) Rotary joint VSWR Checks and power capacity checks (see 3.7 4)
(c) Position error checks and power interruption checks (see 3.6.1.4, 3 6.1 5 and 3 8.5)
(d) Pedestal rotation checks - 2 to 30 rpm when slaved, 15 rpm when free-running (see 3.6.1)
(e) Protective circuit checks (see 3 9, 3 10 and 3 8 2)

At the conclusion of the operating test, the antenna shall be removed and the pedestal group shall be examined for evidence of physical damage caused by the operating test.

4.5.3 Weights and dimensions Weights and dimensions of the pedestal group shall be determined in accordance with MIL-E-16400 to determine compliance with 3 5.14 In addition, the dimensions listed below shall be gaged or measured to determine compliance with the requirements of 3.7.5.

(a) Antenna mounting hole pattern and alignment hole locations
(b) Alignment hole and mounting hole diameter

4.5 4 Supply line voltage and frequency The pedestal group shall be subjected to supply line voltage and frequency tests in accordance with MIL-E-16400 using voltages and frequency specified in 3.5.15 (Exception: transient frequency test is not required).

4 5 5 Enclosure test The following enclosure tests shall be conducted in accordance with MIL-E-16400 to determine compliance with 3 5.4

(a) Above deck units - Watertight (per MIL-STD-108)
(b) Below deck units - Dripproof (per MIL-STD-108)

4.5.6 Temperature tests.

4 5 6 1 Non-operating condition Non-operating condition temperature tests shall be conducted on the complete equipment in accordance with MIL-E-16400 to show compliance with 3.5 18

4.5.6.2 Operating condition The following temperature operating tests shall be conducted in accordance with MIL-E-16400 to show compliance with 3 5 18:

(a) Above deck units - Class 2
(b) Below deck units - Class 4

4 5.7 Humidity test Humidity test shall be performed on the complete equipment in accordance with MIL-E-16400. Operating characteristics listed in 4 5.2 shall be measured prior to and after temperature cycling tests and during the humidity test Hz At the conclusion of the tests, the equipment shall be subjected to a visual examination There shall be no mechanical deterioration, as evidenced by corrosion of metal parts or binding of rotating parts, and there shall be no degradation in performance from that obtained prior to the humidity tests

4 5 8 Shock, vibration and inclination The antenna pedestal group (with a government furnished AS-2188 IFF interrogator antenna assembly installed) shall be subjected to the shock, vibration and inclination tests in accordance with MIL-E-16400 to demonstrate compliance with 3 5.3 The equipment shall be turned on and operating during the tests and shall perform all of its functions within the limits specified in this specification at the completion of the tests The method of mounting the units for testing shall be subject to the approval of the command or agency concerned.

4 5 9 Wind speed and icing tests The contractor shall devise a plan for testing the operation of the pedestal group to demonstrate compliance with 3.6.1.1 and 3 6.2. The contractor's test plan shall be submitted for approval to the command or agency concerned

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4 6 Inspection of preparation for delivery Sample packages and packs and the inspection of the preservation and packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation and packaging, packing, and marking. Unless otherwise specified, preparation for delivery shall be in accordance with MIL-E-17555 (see 6.2.1).

6 ·NOTES

6.1 Intended use. This specification is intended for use in the design and construction of a lightweight Antenna Pedestal Group AN/UPA-57

6.2 Ordering data Procurement documents should specify the following.

6.2.1 Procurement requirements.

(a) Title, number and date of this specification
(b) Number of first article samples to be submitted if other than specified in 4 3
(c) Marking, levels of preservation and packaging and packing required (see 5 1)

6 2 2 Contract data requirements Data generated by this document is not deliverable unless specified on the Contract Data Requirements List (DD Form 1423) The data required by this specification includes, but is not restricted to the following.

(a) Reliability program plan (see 3.4.2).
(b) Reliability prediction (see 3.4.3).
(c) Source of failure rates data (see 3.4.4)
(d) Reliability reports (see 3.4 5).
(e) Quality program plan (see 4 1 1)
(f) Test procedures (see 4.1.3).
(g) First article inspection report (see 4.3.1).

6.3 First article. Invitations for bids should provide that the Government reserves the right to waive the requirement for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

Preparing activity
Navy - EC

(Project 5985-N334)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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DEPARTMENT OF THE NAVY
NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

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